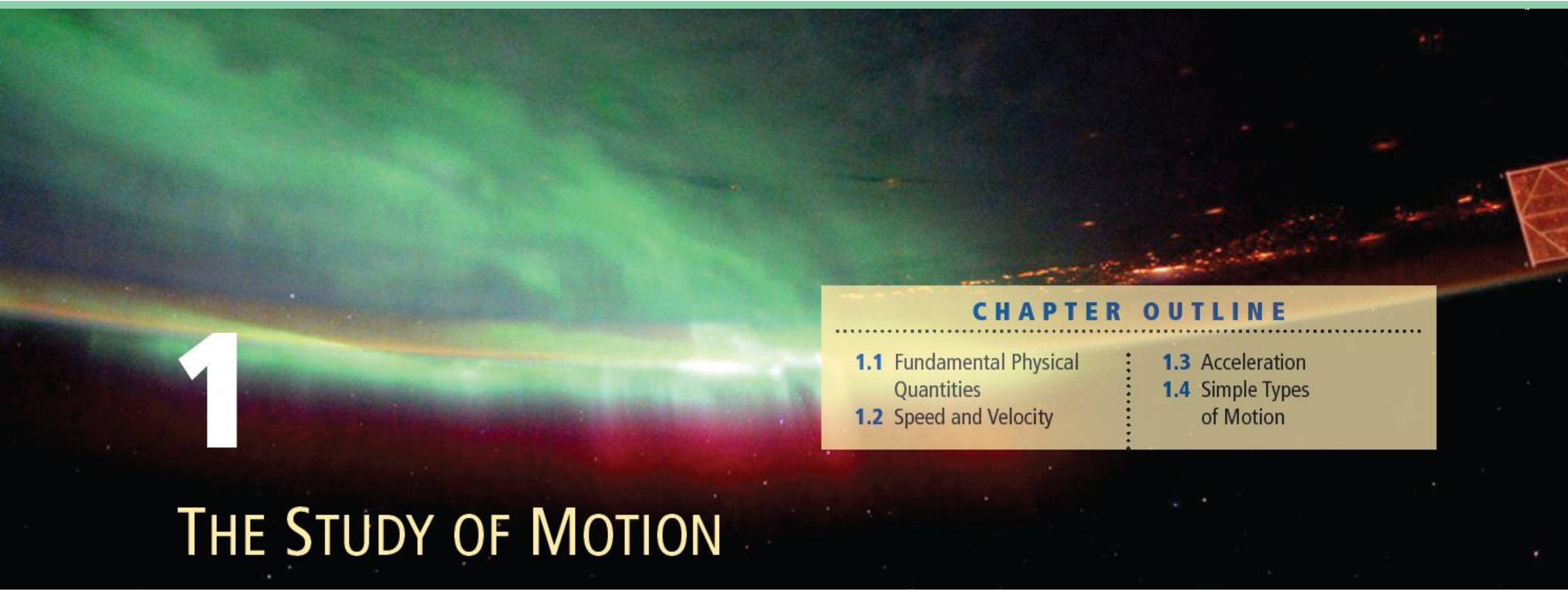

1 The Study of Motion

CHAPTER OUTLINE

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CENGAGE

1.1 Fundamental Physical Quantities

- In physics, there are three basic aspects of the material universe that we must describe and quantify in various ways:

  space, time, and *matter.*

- All physical quantities used here involve measurements (or combinations of measurements) of space, time, and the properties of matter.

- The units of measure of all of these quantities can be traced back to the units of measure of distance, time, and two properties of matter called *mass* and *charge.*

1.1 Fundamental Physical Quantities : Distance

Mostly all quantities can be classified in terms of the fundamental physical quantities:

- Distance (L)
- Mass (M)
- Time (T)

For example, speed has the unit of miles per hour (L/T)

1.1 Fundamental Physical Quantities: Distance

- **Distance** represents a measure of space in one dimension.
  Length, width, and height are examples of distance measurements

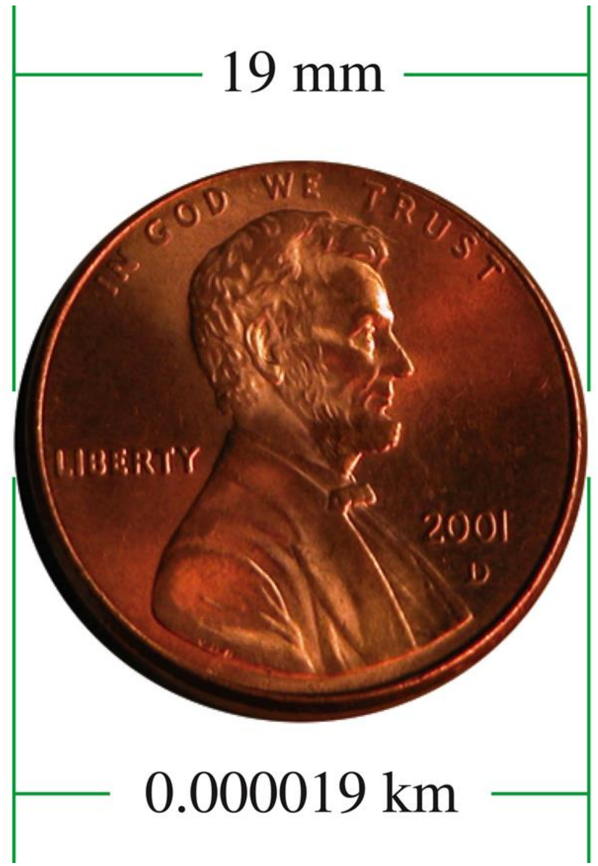
- The following table lists the common distance units of measure, in both the *metric and English systems* and their abbreviations.

Physical Quantity	**Metric Units**	**English Units**
Distance *d* (or *l*, *w*, *h*)	meter (m)	foot (ft)
	millimeter (mm)	inch (in.)
	centimeter (cm)	mile (mi)
	kilometer (km)	

1.1 Fundamental Physical Quantities

Why are there so many different units in each system?

Generally, it is easier to use a unit that fits the scale of the system being considered. The meter is good for measuring the size of a house, the millimeter for measuring the size of a coin, and the kilometer for measuring the distance between cities

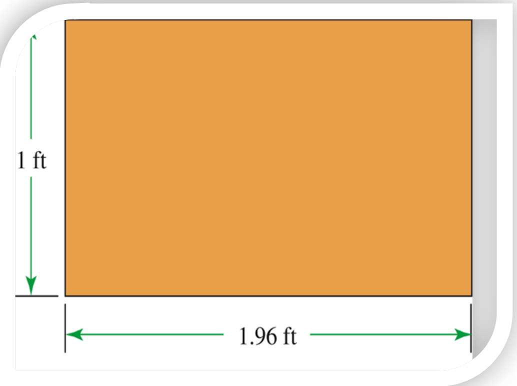


1.1 Fundamental Physical Quantities: Area and Volume

- Two other physical quantities that are closely related to distance are *area* and *volume*.
- Area and volume are examples of physical quantities that are based on other physical quantities—in this case, just distance.
- Area commonly refers to the size of a surface.

Example: the floor in a room or the outer skin of a basketball.

- The concept of area can apply to *surfaces that are not flat* and to *“empty,” two-dimensional spaces* such as holes and open windows
- Every surface, whether it is flat or curved, has an area. The area of this rectangle and the area of the surface of the basketball are equal



1.1 Fundamental Physical Quantities: Area and Volume

The following table lists the common area and volume units of measure, in both the *metric and English systems* and their abbreviations.

Physical Quantity	**Metric Units**	**English Units**
Area (A)	square meter (m^2)	square foot (ft^2)
	square centimeter (cm^2)	square inch ($in.^2$)
	square kilometer (km^2)	square mile (mi^2)
	hectare	acre
Volume (V)	cubic meter (m^3)	cubic foot (ft^3)
	cubic centimeter (cm^3 or cc)	cubic inch ($in.^3$)
	liter (L)	quart, pint, cup
	milliliter (mL)	teaspoon, tablespoon

1.1 Fundamental Physical Quantities: Converting Distance Measurements

Example: Convert 23 meters to feet.

$$23 \text{ meters} = 23 \times 1 \text{ meter}$$

$$1 \text{ meter} = 3.28 \text{ feet:}$$

$$23 \text{ meters} = 23 \times 1 \text{ meter} = 23 \times 3.28 \text{ feet}$$

$$23 \text{ meters} = 75.4 \text{ feet}$$

1.1 Fundamental Physical Quantities: Time

- The measure of **time** is based on periodic phenomena—processes that repeat over and over at a regular rate.

- Both the *metric system* and the *English system* use the same units for time

Physical Quantity	**Metric Units**	**English Units**
Time (*t*)	second (s)	second (s)
	minute (min)	minute (min)
	hour (h)	hour (h)

1.1 Fundamental Physical Quantities: Time

DEFINITION

Period: The time for one complete cycle of a process that repeats. It is abbreviated T, and the units are seconds, minutes, and so forth.

DEFINITION

Frequency: The number of cycles of a periodic process that occur per unit time. It is abbreviated f and has units s^{-1} or hertz (Hz).

1.1 Fundamental Physical Quantities: Time

The relationship between the period of a cyclic phenomenon and its frequency is simple.

The period equals 1 divided by the frequency, and vice versa.

$$\text{period} = \frac{1}{\text{frequency}}$$

$$T = \frac{1}{f} \rightarrow f = \frac{1}{T}$$

The standard unit of frequency is the hertz (Hz), which equals 1 cycle per second.

$$1 \text{ Hz} = 1/\text{s} = 1 \text{ s}^{-1}$$

$$91.5 \text{ MHz} = 91{,}500{,}000 \text{ Hz}$$

$$2 \text{ GHz} = 2{,}000{,}000{,}000 \text{ Hz}$$

Example 1.1

A mechanical stopwatch uses a balance wheel that rotates back and forth 10 times in 2 seconds. What is the frequency of the balance wheel?

$$\text{frequency} = \text{number of cycles per time}$$

$$f = \frac{10 \text{ cycles}}{2 \text{ s}} = 5 \text{ Hz}$$

What is the period of the balance wheel?

$$\begin{aligned} \text{period} &= \text{time for one cycle} \\ &= 1 \text{ divided by the frequency} \end{aligned}$$

$$T = \frac{1}{5 \text{ Hz}} = 0.2 \text{ s}$$

1.1 Fundamental Physical Quantities: Mass

- Mass is also a measure of what we sometimes refer to in everyday speech as *inertia*.

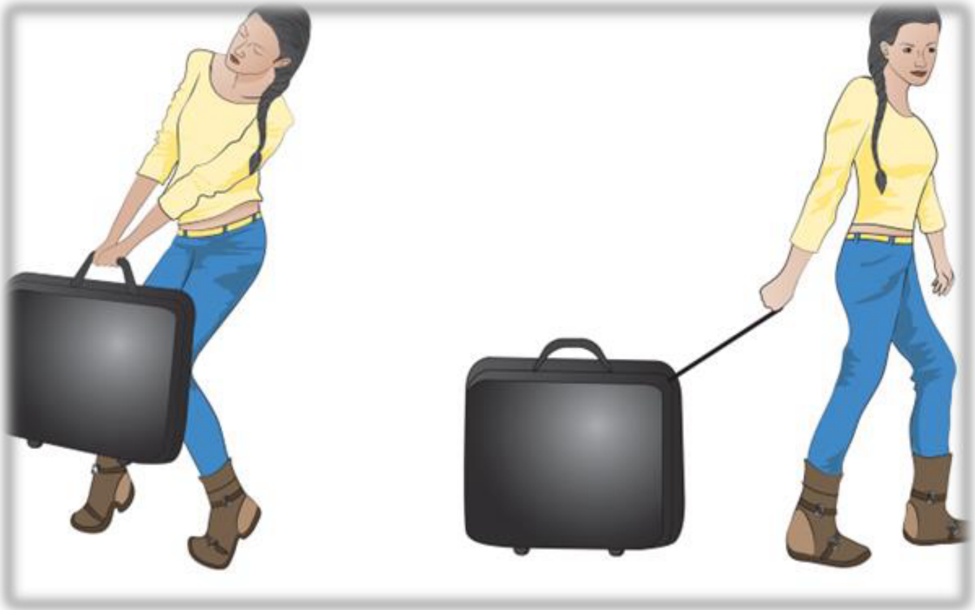
  The larger the mass of an object, the greater its inertia and the more difficult it is to speed up or slow down.

- Mass is not in common use in the English system; note the unfamiliar unit, the slug.
- The following table lists the mass unit of measure in the *metric and English systems* and their abbreviations.

Physical Quantity	Metric Units	English Units
Mass (m)	kilogram (kg)	slug
	gram (g)	

1.1 Fundamental Physical Quantities: Mass

- Weight, a quantity that is related to, but is *not* the same as mass, is used instead.
- When you try to lift a heavy suitcase, you are experiencing its weight.
- When you pull it along and speed it up or slow it down, you are experiencing its mass.



1.2 Speed and Velocity

A key concept to use when quantifying motion is **speed**.

DEFINITION

Speed: Rate of movement. Time rate of change of distance from a reference point. The distance traveled divided by the time elapsed.

It is important to distinguish between *average speed* and *instantaneous speed*.

1.2 Speed and Velocity

An object's average speed is the total distance it travels during some period of time divided by the time that elapses.

$$\text{average speed} = \frac{\text{total distance}}{\text{total elapsed time}}$$

The following table lists the mass unit of measure, in both the *metric and English systems* and their abbreviations.

Physical Quantity	**Metric Units**	**English Units**
Speed (v)	meter per second (m/s)	foot per second (ft/s)
	kilometer per hour (km/h)	mile per hour (mph)

1.2 Speed and Velocity

In some cases, an object may have been traveling for a while and already moved some distance before we start taking measurements to determine its speed (average or instantaneous).

Then the distance and time that we use would be the values at the end of the segment being timed (the *final* values) minus the values at the beginning of the segment (the *initial* values)—that is, the *changes* in distance and time.

In mathematical notation,

$$\text{total distance} = d_{\text{final}} - d_{\text{initial}} = d_f - d_i$$

$$\text{total elapsed time} = t_{\text{final}} - t_{\text{initial}} = t_f - t_i$$

1.2 Speed and Velocity

The general expression for speed is:

$$v = \frac{d_{\text{final}} - d_{\text{initial}}}{t_{\text{final}} - t_{\text{initial}}} = \frac{d_f - d_i}{t_f - t_i}$$

$$= \frac{\Delta d}{\Delta t}$$

The symbol Δ is the Greek letter delta and represents the *change in*.

1.2 Speed and Velocity

As the time interval becomes shorter and shorter, we approach the *instantaneous speed*.

Instantaneous speed is the speed that an object has at an instant in time.

$$\text{instantaneous speed} \approx \frac{\text{very short distance}}{\text{very short time}}$$

1.2 Speed and Velocity

For example, given sophisticated equipment, we might measure how long it takes a car to travel 1 meter. If that time is found to be 0.05 seconds (not an "instant" but a very short time), a good estimate of the car's instantaneous speed is

Example:

$$\text{instantaneous speed} \approx \frac{1\ \text{m}}{0.05\ \text{s}}$$
$$= 20\ \text{m/s} = 44.8\ \text{mph}$$

$$20\ \text{m/s} = 20 \times 1\ \text{m/s} = 20 \times 2.24\ \text{mph}$$
$$= 44.8\ \text{mph}$$

Example 1.2

An analysis of a videotape of Olympic gold-medal winner Usain Bolt running the 100-meter dash in the 2009 in Berlin yields the data in the table. Compute his average speed for the race and estimate his peak instantaneous speed.

$$\text{average speed} = v = \frac{\Delta d}{\Delta t}$$

$$= \frac{d_{\text{final}} - d_{\text{initial}}}{t_{\text{final}} - t_{\text{initial}}}$$

$$v = \frac{100\text{ m} - 0\text{ m}}{9.69\text{ s} - 0\text{ s}} = \frac{100\text{ m}}{9.69\text{ s}}$$

$$= 10.32\text{ m/s } (= 23.1\text{ mph})$$

Segment (meters)	**Time (seconds)**
0 – 10	1.85
10 – 20	1.02
20 – 30	0.91
30 – 40	0.87
40 – 50	0.85
50 – 60	0.82
60 – 70	0.82
70 – 80	0.82
80 – 90	0.83
90 – 100	0.90
Total Distance: 100 m	Total Time: 9.69 s

Example 1.2, continued

The nearly equally spaced times between 50 and 90 meters indicate that his speed was constant. Therefore, we can use any segment from this part of the race to compute his instantaneous speed.

$$v = \frac{\Delta d}{\Delta t} = \frac{d_{\text{final}} - d_{\text{initial}}}{t_{\text{final}} - t_{\text{initial}}}$$

$$= \frac{80\text{ m} - 70\text{ m}}{0.82\text{ s}}$$

$$= \frac{10\text{ m}}{0.82\text{ s}} = 12.2\text{ m/s}$$

$$= 27.3\text{ mph}$$

Segment (meters)	**Time (seconds)**
0 – 10	1.85
10 – 20	1.02
20 – 30	0.91
30 – 40	0.87
40 – 50	0.85
50 – 60	0.82
60 – 70	0.82
70 – 80	0.82
80 – 90	0.83
90 – 100	0.90
Total Distance: 100 m	Total Time: 9.69 s

1.2 Speed and Velocity

When the speed of an object is constant, the average and instantaneous speeds are the same.

In this case, we can express the relationship between the distance travelled and the time that has elapsed as follows:

$$d = vt \quad (\text{when speed is constant})$$

This is an example of what is called *direct proportionality*.

The distance is *proportional* to the elapsed time: $d \propto t$

Using the speed gives us equality, i.e., an equal sign, so we call v the *proportionality constant*.

1.2 Speed and Velocity

An important aspect of the motion is direction.

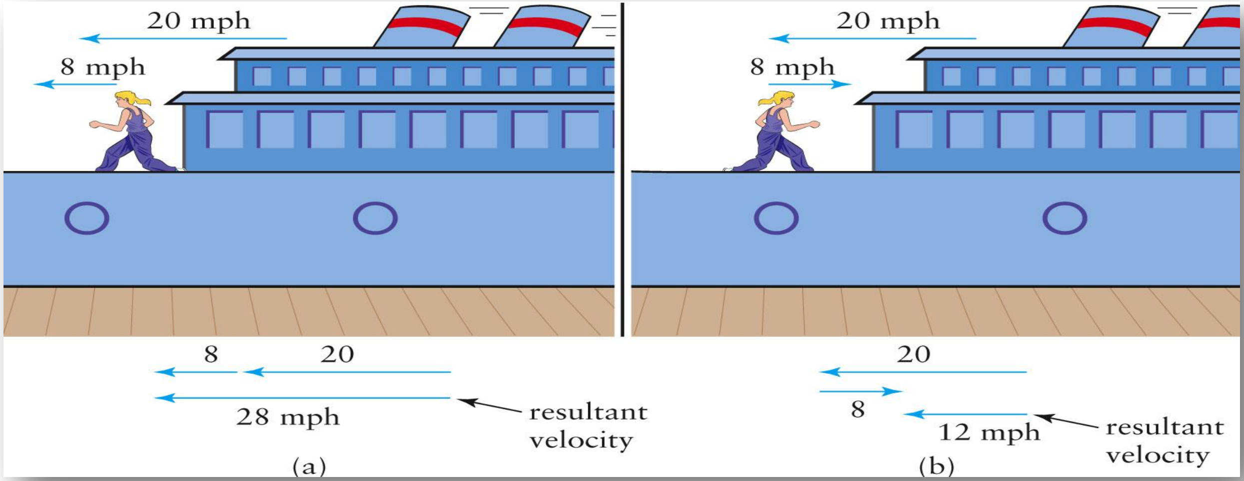
DEFINITION

Velocity Speed in a particular direction (same units as speed). Directed motion.

Velocity is an example of a physical quantity called a **vector**.

Speed by itself is a scalar.

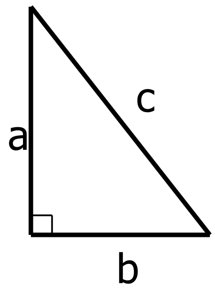
1.2 Speed and Velocity: Vector Addition



- Quantities that convey a magnitude and a direction, like velocity, are called **vectors.**
- We represent vectors by an arrow; its length indicates the magnitude.

1.2 Speed and Velocity: Vector Addition

- *Pythagorean theorem*.



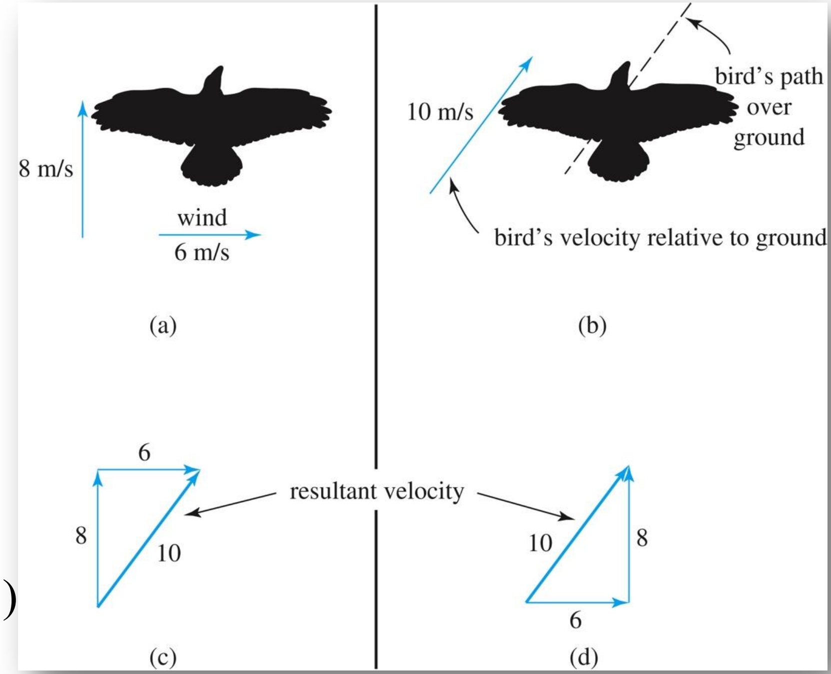
$$c^2 = a^2 + b^2$$

$$c = \sqrt{a^2 + b^2}$$

- Let's find the net speed of the bird. (Why didn't I say net velocity?)

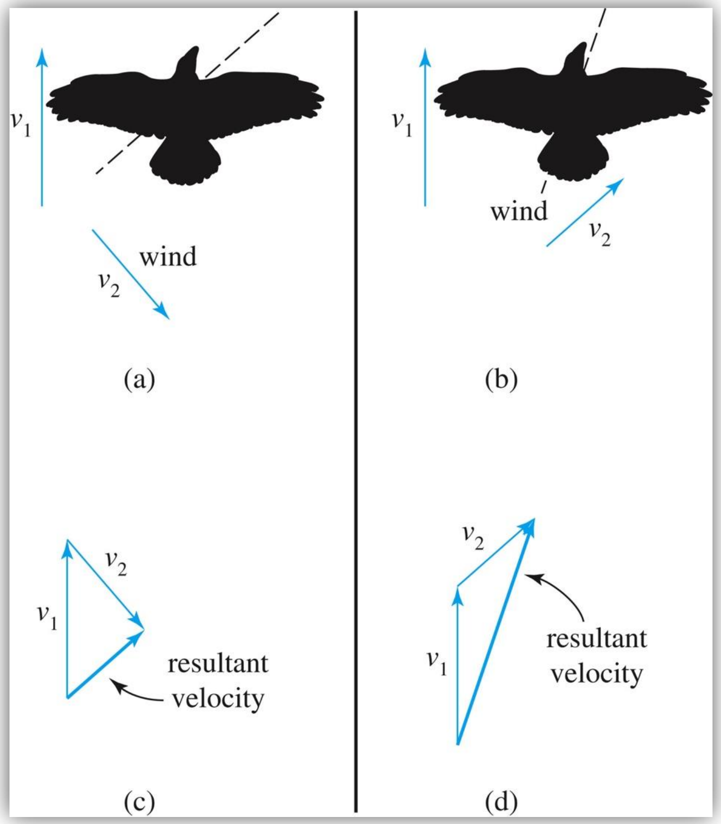
$$\sqrt{6^2 + 8^2} = \sqrt{100}$$

$$= 10$$



1.2 Speed and Velocity: Vector Addition

The bird has the same speed and direction in the air, but the wind direction is different.



1.3 Acceleration

DEFINITION

Acceleration: Rate of change of velocity; the change in velocity divided by the time elapsed:

$$a = \Delta v / \Delta t$$

The following table lists the acceleration unit of measure, in both the *metric and English systems* and their abbreviations.

Physical Quantity	Metric Units	English Units
Acceleration (a)	meter per second^2 (m/s^2)	foot per second^2 (ft/s^2)
		mph per second (mph/s)

1.3 Acceleration

- A common way to express acceleration is in terms of g's.

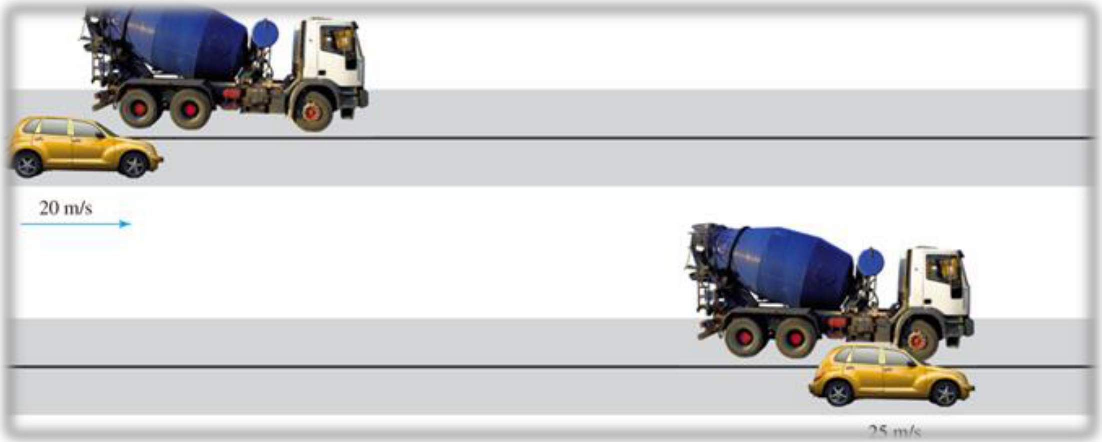
- One g is the acceleration an object experiences as it falls near the Earth's surface: $g = 9.8\ \text{m/s}^2$.

if you experience 2g during a collision, your acceleration was 19.6 m/s².

- Since velocity is speed and direction, there are three ways it can change:
  - change in speed,
  - change in direction, or
  - change in both speed and direction.
- The change in direction is an important case often misunderstood

Example 1.3

A car accelerates from 20 to 25 m/s in 4 seconds as it passes a truck. What is its acceleration?



$$a = \frac{\Delta v}{\Delta t} = \frac{\text{final speed} - \text{initial speed}}{\Delta t}$$

$$= \frac{25\text{ m/s} - 20\text{ m/s}}{4\text{ s}} = 1.25\text{ m/s}^2$$

Example 1.4

After a race, a runner, traveling in a fixed positive direction, takes 5 seconds to come to a stop from a speed of 9 m/s. The acceleration is as follows:

$$a = \frac{\Delta v}{\Delta t} = \frac{0\text{ m/s} - 9\text{ m/s}}{5\text{ s}} = \frac{-9\text{ m/s}}{5\text{ s}} = -1.8\text{ m/s}^2$$

What's up with the minus sign?

1.3 Acceleration

Perhaps the most important example of accelerated motion is that of an object falling freely near the Earth's surface.

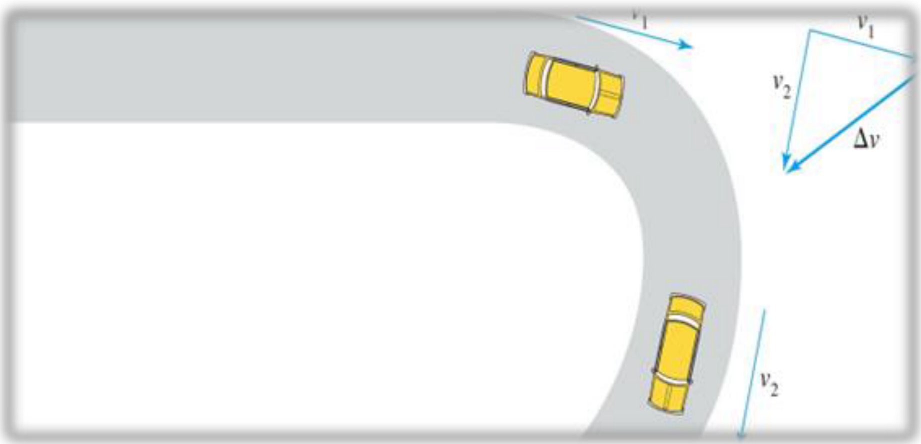
Free fall: Only the force of gravity is acting, and we ignore things like air resistance.

A rock falling a few meters would satisfy this condition, but a feather would not.

1.3 Acceleration: Centripetal Acceleration

Imagine a car rounding a curve.

The car’s velocity must keep changing toward the center of the curve in order to stay on the road.



So, there is an acceleration toward the center of the curve.

Centripetal means “center-seeking.”

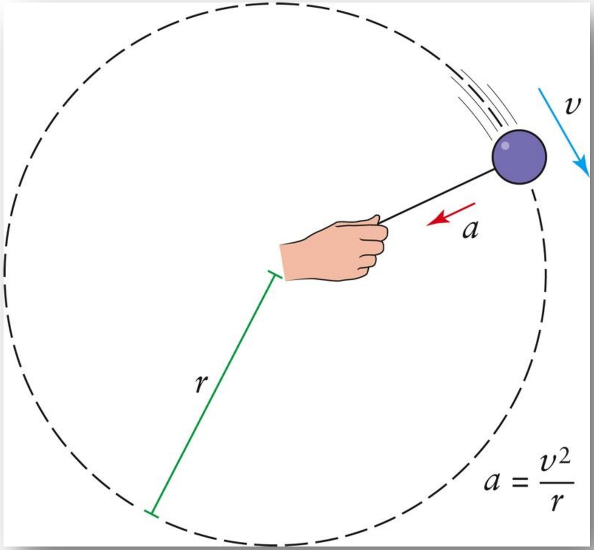
1.3 Acceleration: Centripetal Acceleration

Centripetal acceleration: is the acceleration associated with an object moving in a circular path.

$$a = \frac{v^2}{r} \quad (\text{centripetal acceleration})$$

$$a \propto v^2$$

$$a \propto \frac{1}{r}$$

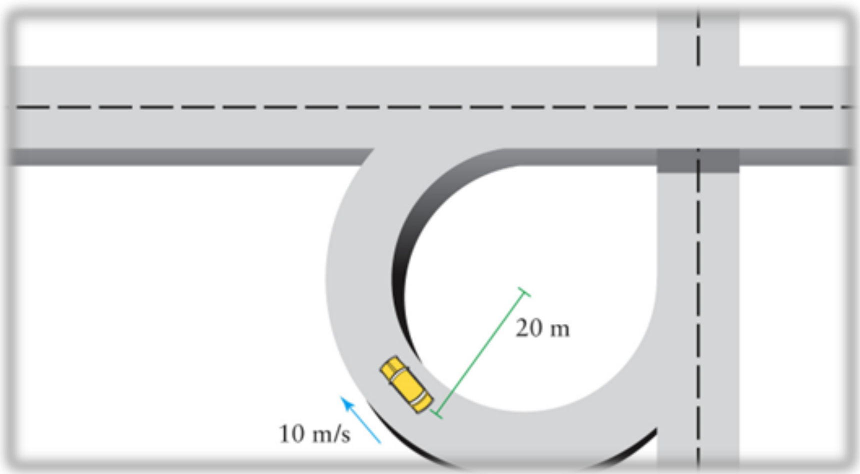


Example 1.5

Let's estimate the acceleration of a car as it goes around a curve. The radius of a segment of a typical cloverleaf highway interchange is 20 meters, and a car might take the curve with a constant speed of 10 m/s (about 22 mph).

$$a = \frac{v^2}{r} = \frac{(10\ \text{m/s})^2}{20\ \text{m}}$$

$$= \frac{100\ \text{m}^2/\text{s}^2}{20\ \text{m}} = 5\ \text{m/s}^2$$



1.4 Simple types of motion: Constant Velocity

The simplest type of motion is obviously ***no motion***: a single body resting at a fixed position in space. The object has no velocity.

The object's position, relative to some reference, is constant. This means that the object's velocity (and acceleration) are zero.

1.4 Simple types of motion: Constant Velocity

- The next simplest type of motion is ***uniform motion***.
- The object's moves with a constant non-zero velocity; that is, with a constant speed in a fixed direction.
- The position of the object, relative to some reference, is proportional to time.
- Example: an automobile travelling on a straight, flat highway at a constant speed.

1.4 Simple types of motion: Constant Velocity

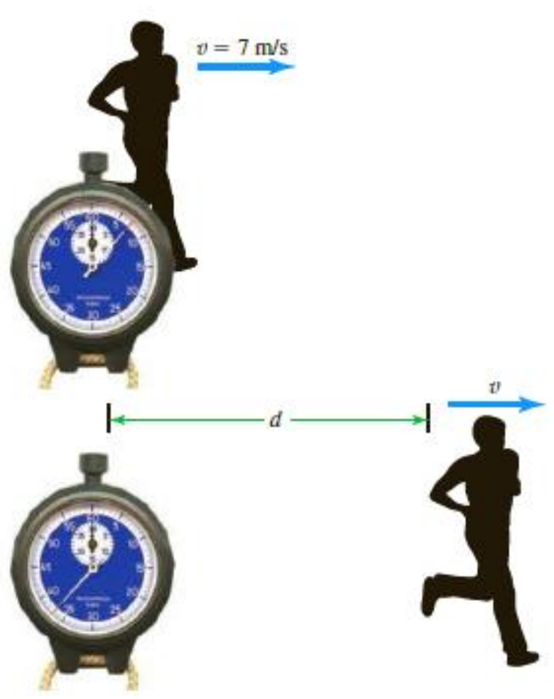
The relationship here is between distance and time (no change in velocity, no acceleration)

We can express that relationship in four different ways: with words, mathematics, tables, and graphs.

1.4 Simple types of motion: Constant Velocity

Let's take the example of a runner travelling at a steady pace of 7 m/s.

If you are standing on the side of the road, how does the distance from you to the runner change with time after the runner has gone past you?



In words, the distance increases by 7 meters each second: two seconds after the runner passes by, the distance would be 14 meters.

1.4 Simple types of motion: Constant Velocity

Mathematically

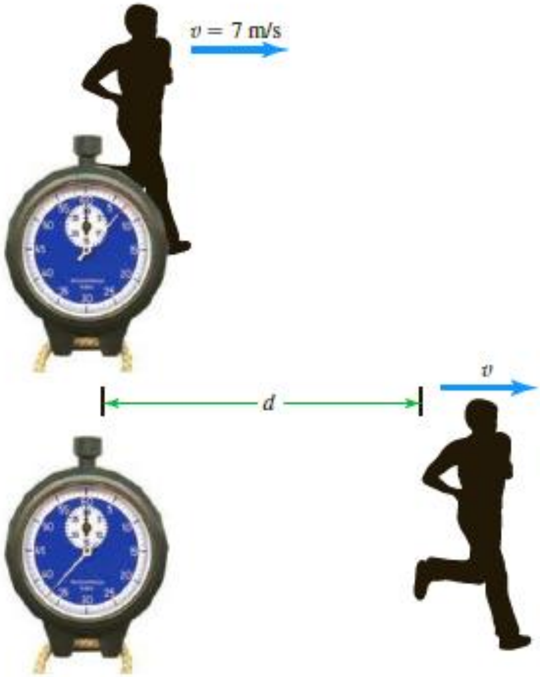
$$d = 7t$$

$$(d \text{ in meters, } t \text{ in seconds})$$

This is an example of the general equation:

$$d = vt$$

(with $v = 7$ m/s)



1.4 Simple types of motion: Constant Velocity

table

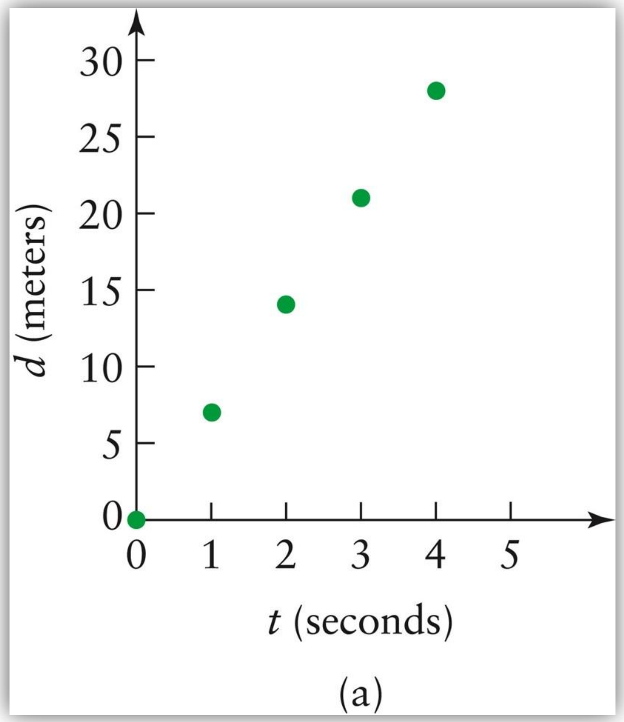
- The same information can also be put in a table of values of time and distance.
- The accompanying table shows the distance (d) values at certain times (t). These values all satisfy the equation $d = 7t$.

Time (s)	Distance (m)
0	0
1	7
2	14
3	21
4	28

1.4 Simple types of motion: Constant Velocity

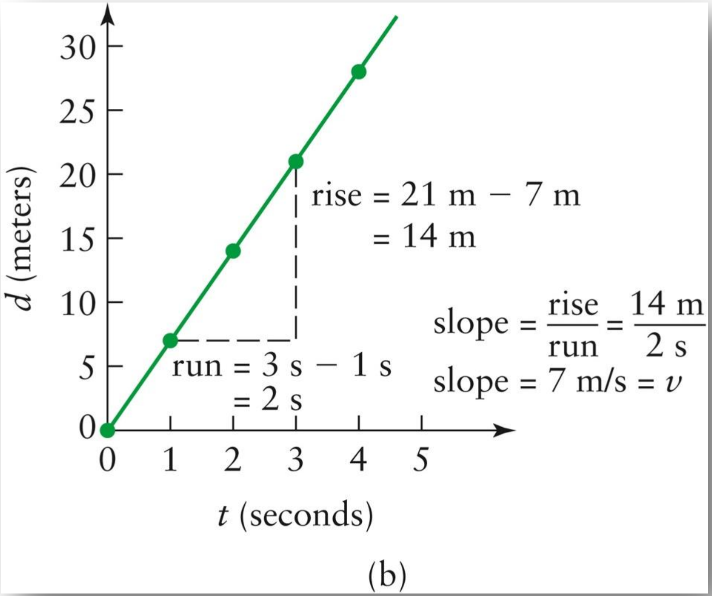
Graph

The fourth way to show the relationship between distance and time is to *graph* the values in the table.
The usual practice is to graph distance versus time, which puts distance on the vertical axis.



1.4 Simple types of motion: Constant Velocity

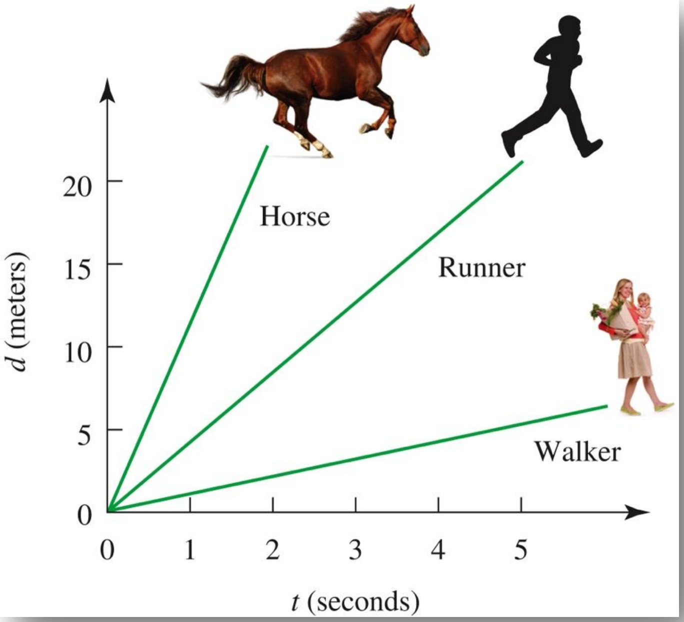
- If the speed is constant, the distance versus time graph is a straight line.
- An important feature of this graph is its *slope*.
- The slope of a graph is a measure of its steepness.
- In particular, the slope is equal to the *rise* between two points on the line divided by the *run* between the points.
- The slope of a distance-versus-time graph equals the velocity.



1.4 Simple types of motion: Constant Velocity

- The graph of *d* versus *t* for a faster-moving body, a racehorse, for instance, would be steeper—it would have a larger slope.

- The graph of *d* versus *t* for a slower object (a person walking) would have a smaller slope.

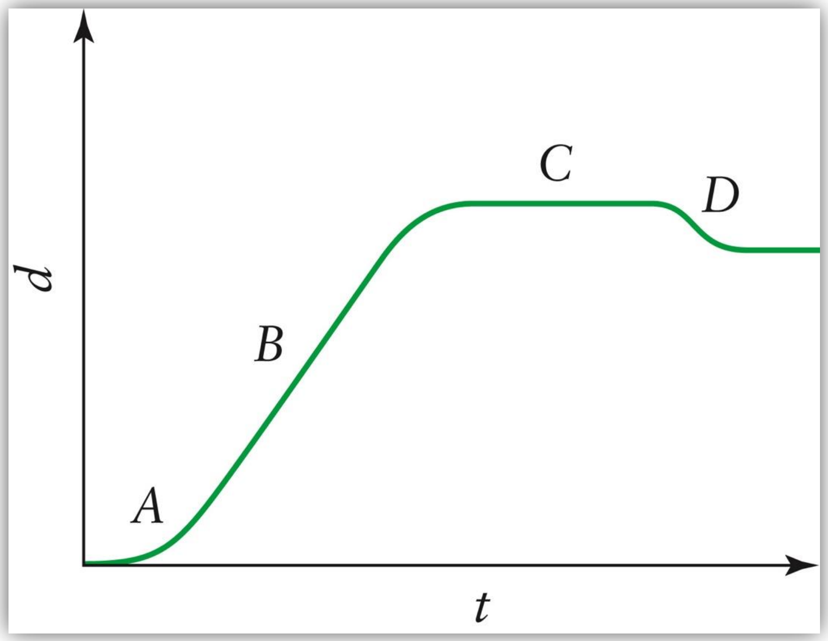


1.4 Simple types of motion: Constant Velocity

When an object stands still (without motion), the graph of d versus t is a flat, horizontal line parallel to the time axis. The slope is zero because the velocity is zero.

Even when the velocity is not constant, the slope of a d versus t graph still equals the velocity. In this case, the graph is not a straight line because the slope changes.

1.4 Simple types of motion: Constant Velocity



The accompanying represents the motion of a car that starts from a stop sign (A), drives down a street (B), and then stops and backs into a parking place (C). The car is backing up, and the graph is slanted downward. The distance is decreasing, and the velocity is negative(D).

1.4 Simple types of motion: Constant Acceleration

- The next example of simple motion is ***constant acceleration in a straight line.***

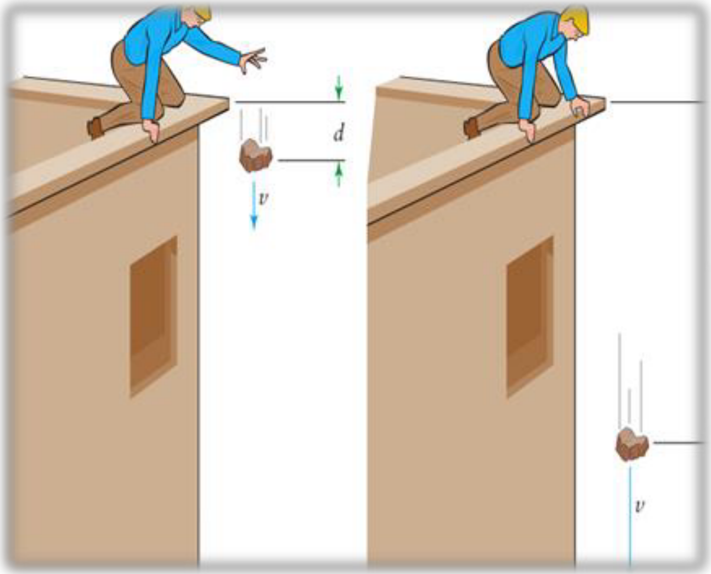
- The object's velocity is changing at a fixed rate.

- Examples of object motions with constant acceleration in a straight line :
  - A freely falling body.
  - A ball rolling down a straight, inclined plane.

1.4 Simple types of motion: Constant Acceleration

Assume that a heavy rock is dropped from the top of a building.

- The rock falls with an acceleration equal to *g*.
- This is 9.8 m/s2, which is equivalent to 22 mph per second.
- The velocity increases by 9.8 m/s, or 22 mph, each second.
- This means that the rock's velocity equals



$$v = 9.8t \quad (v \text{ in m/s}, t \text{ in seconds})$$

$$v = 22t \quad (v \text{ in mph}, t \text{ in seconds})$$

1.4 Simple types of motion: Constant Acceleration

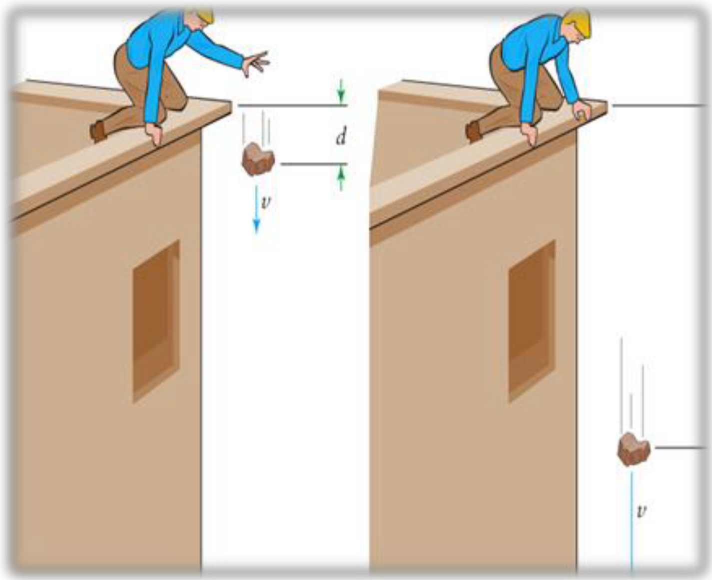
Assume that a heavy rock is dropped from the top of a building.

For any object that starts from rest and has a constant acceleration a is:

$$v = at \quad (\text{when acceleration is constant})$$

In constant acceleration, the velocity is proportional to the time.

The proportionality constant is the acceleration a.



1.4 Simple types of motion: Constant Acceleration

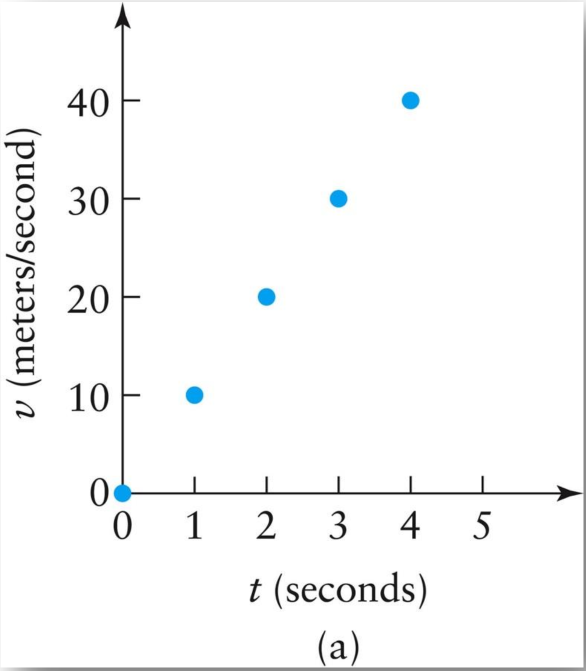
The accompanying table gives the relevant values for this example

Time	Velocity	
(s)	(m/s)	(mph)
0	0	0
1	9.8	22
2	19.6	44
3	29.4	66
4	39.2	88

1.4 Simple types of motion: Constant Acceleration

- The graph of velocity versus time is a straight line.
- The slope of a graph of velocity versus time equals the acceleration.
- The rise is a change in velocity, Δv, and the run is a change in time Δt.

$$\text{slope} = \frac{\Delta v}{\Delta t} = a$$



1.4 Simple types of motion: Constant Acceleration

If an object is already moving with velocity v_{initial} and then undergoes constant acceleration, its velocity after time t is:

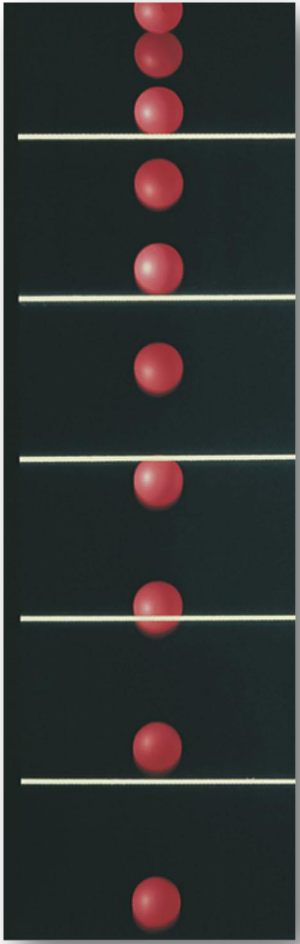
$$v = v_{\text{initial}} + at.$$

Clearly, in the case of an object starting from rest, v_{initial} is zero.

$$v = at$$

1.4 Simple types of motion: Constant Acceleration

What is the relationship between distance and time when the acceleration is constant?



The distance a falling body travels during each successive time interval grows larger as it falls. Because the velocity is continually changing, the distance equals the average velocity multiplied by the time.

The object starts with velocity equal to zero; after accelerating for a time *t*, its velocity is *at*

1.4 Simple types of motion: Constant Acceleration

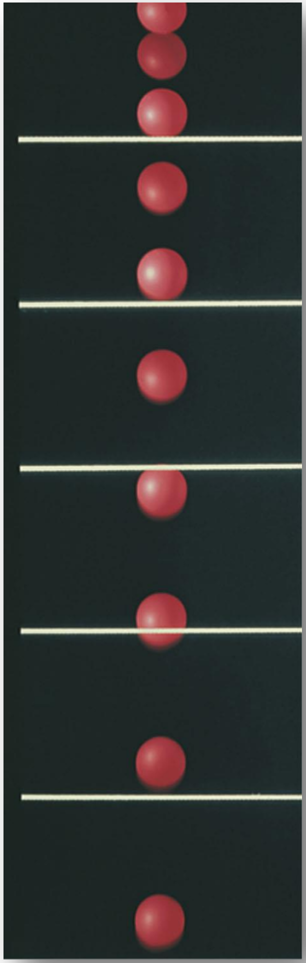
What is the average velocity?

$$\text{average velocity} = \frac{0 + at}{2} = \frac{1}{2}at$$

The distance traveled is this average velocity times the time.

$$d = \text{average velocity} \times t = \frac{1}{2}at \times t$$

$$d = \frac{1}{2}at^2 \quad \left(\begin{array}{l} \text{when initial velocity is zero} \\ \text{and acceleration is constant} \end{array} \right)$$



1.4 Simple types of motion: Constant Acceleration

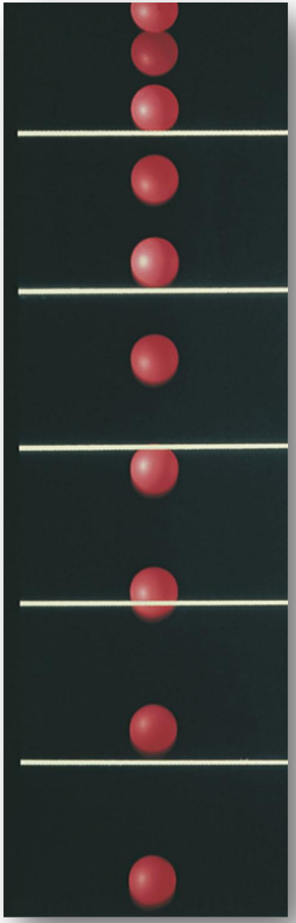
In the case of a falling body, the acceleration is 9.8 m/s^2; therefore,

$$d = \frac{1}{2}at^2 = \frac{1}{2} \times 9.8 \times t^2$$

$$= 4.9t^2 \quad (d \text{ in meters, } t \text{ in seconds})$$

In the case of *constant acceleration*, the distance is proportional to the square of the time.

The constant of proportionality is one-half the acceleration.



1.4 Simple types of motion: Constant Acceleration

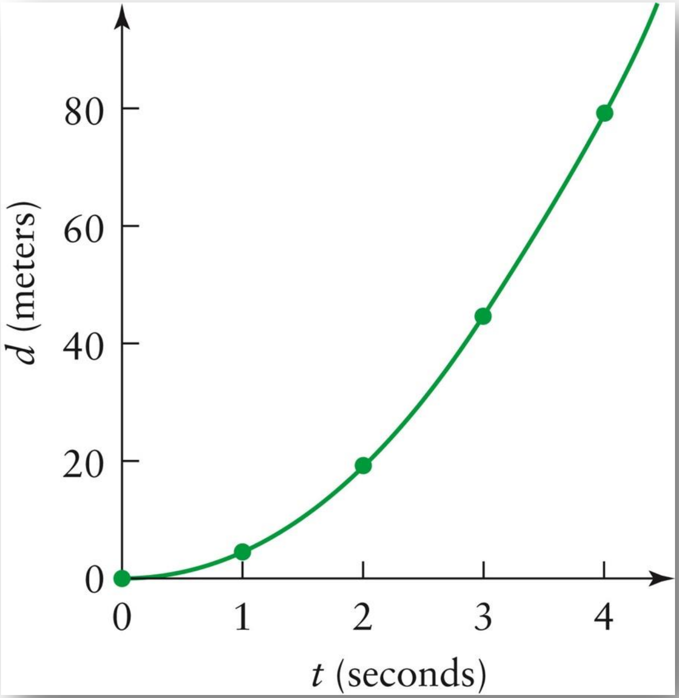
For any object that is already moving with velocity *v* initial and then undergoes constant acceleration, the average velocity after time *t* is:

$$v_{\text{average}} = \frac{v_{\text{initial}} + (v_{\text{initial}} + at)}{2} = v_{\text{initial}} + \frac{1}{2}at$$

$$d = v_{\text{average}}t = v_{\text{initial}}t + \frac{1}{2}at^2$$

1.4 Simple types of motion: Constant Acceleration

- The distance versus time graph for a freely falling body curves upward.
- The increasing slope indicates that the velocity increases throughout the motion.
- The body's velocity increases with time, and the slope of this graph equals the velocity.



1.4 Simple types of motion: Constant Acceleration

Constant Acceleration

Type of Motion	**Behavior of Physical Quantities**	**Equations**
Stationary object	Distance constant	d = constant
	Velocity zero	$v = 0$
	Acceleration zero	$a = 0$
Uniform motion*	Distance proportional to time	$d = vt$
	Velocity constant	v = constant
	Acceleration zero	$a = 0$
Uniform acceleration* (from rest)	Distance proportional to time squared	$d = \frac{1}{2} at^2$
	Velocity proportional to time	$v = at$
	Acceleration constant	a = constant
*Distance measured from object's initial location.		

Important Equations

IMPORTANT EQUATIONS

Equation	Comments	Equation	Comments
Fundamental Equations		*Special-Case Equations*	
$T = \frac{1}{f}$	Relates period and frequency	$d = vt$	Distance from starting point when velocity is constant
$f = \frac{1}{T}$	Relates frequency and period	$d = \frac{1}{2} at^2$	Distance when acceleration is constant and object starts from rest
$v = \frac{\Delta d}{\Delta t}$	Definition of velocity or speed	$v = at$	Velocity when acceleration is constant and object starts from rest
$a = \frac{\Delta v}{\Delta t}$	Definition of acceleration	$a = g$	Acceleration of freely falling body
		$a = \frac{v^2}{r}$	Centripetal acceleration (object moving in a circular path)